

Vector Spaces

Vector Spaces

Defn. A **vector space** is a collection of objects (called **vectors**) with operations addition and scalar multiplication that obey the “usual” vector laws.

That is, addition and scalar multiplication behave like they do for ordinary vectors.

Overview of The Axioms of a Vector Space

- the space is closed under addition and scalar multiplication
- addition is **commutative** (order doesn't matter), **associative** (brackets don't matter), and has **negation**;
- the 0 vector and 1 scalar behave as **identities**; and
- addition and scalar multiplication **distribute** (interact nicely).

Axioms of a Vector Space

For all vectors u , v , and w in V and all (real) scalars c and d :

1) The sum $u + v$ is in V

2) $u + v = v + u$

3) $(u + v) + w = u + (v + w)$

4) There is a vector 0 such that $u + 0 = u$

5) There is a vector $-u$ such that $u + (-u) = 0$

Axioms of a Vector Space Continued

6) The scalar multiple cu is in V

$$7) c(u + v) = cu + cv$$

$$8) (c + d)u = cu + du$$

$$9) c(du) = (cd)u$$

$$10) 1u = u$$

Example $\mathbb{R}^n$

$\mathbb{R}^n$, with addition and scalar multiplication as we've been doing, is a vector space.

Example Vector Spaces

Defn. $\mathbb{P}_n$ is the set of all polynomials of degree at most n ; $\mathbb{P}$ is the set of all polynomials.

Defn. $C[t]$ is the set of all continuous functions in variable t with domain $\mathbb{R}$

Defn. M_n is the set of all $n \times n$ matrices.

Definitions

The earlier definitions carry over to abstract vector spaces:

- + linear combination
- + span
- + subspace
- + linearly independent
- + basis
- + dimension

Examples of Subspaces

- ⤿ The degree-bounded polynomial space $\mathbb{P}_n$ is a subspace of the space $\mathbb{P}$ of all polynomials.
- ⤿ $\mathbb{P}$ is a subspace of the space $C[t]$ of continuous functions.
- ⤿ The set of continuous functions such that $\int_{-\infty}^{\infty} f(t) dt = 0$ is a subspace of $C[t]$.

Examples of Dimension

- ⤿ $\mathbb{P}_n$ has dimension $n+1$. A basis is $\{1, t, t^2, \dots, t^n\}$.
- ⤿ The set $C[t]$ of continuous functions has infinite dimension.
- ⤿ The set M_2 of 2×2 matrices has dimension 4. A basis is the quartet of 2×2 matrices with three 0's and one 1.

Summary

A vector space is a collection of objects with addition and scalar multiplication that obey the “usual” vector laws: it is closed under addition and scalar multiplication; addition is commutative, associative and has negation; the 0 vector and 1 scalar act as identities; and addition and scalar multiplication distribute.

Summary (cont)

Examples include $\mathbb{R}^n$; $\mathbb{P}_n$ the set of all polynomials of degree at most n ; $C[t]$ the set of all continuous functions in variable t ; and M_n the set of all $n \times n$ matrices. The concepts of linearly independent, span, subspace, basis, and dimension carry over.